x_1	x_2	x_3	x_4	x_5	x_6
(1)	(2)	(1)	(5)	(6)	(6)
(1)	(1)	(2)	(4)	(5)	(6)
C_1	C_1	C_1	C_2	C_2	C_2

Initial $\underline{W} = \begin{pmatrix} 4 \\ -2 \\ -2 \end{pmatrix}$ **Step η = 2**

Check classes of $\begin{pmatrix} z_1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} z_1 \\ 7 \\ 6 \end{pmatrix} \begin{pmatrix} z_1 \\ 3 \\ 2 \end{pmatrix}$

C_1 positive C_2 negative Zero is unaccepted

Ugmented Vectors (from Top)

x_1	x_2	x_3	x_4	x_5	x_6
1	1	1	1	1	1
1	2	1	5	6	6
1	1	2	4	5	6
C_1	C_1	C_1	C_2	C_2	C_2

Iteration 2

$d(x_1) = \begin{vmatrix} 1 & 1 & 1 \end{vmatrix} \begin{pmatrix} 4 \\ -10 \\ -8 \end{pmatrix} = -14$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 4 \\ -10 \\ -8 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix}$ ⇐

$d(x_2) = \begin{vmatrix} 1 & 2 & 1 \end{vmatrix} \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix} = -14$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 6 \\ -8 \\ -6 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix}$ ⇐

$d(x_3) = \begin{vmatrix} 1 & 1 & 2 \end{vmatrix} \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix} = -4$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 8 \\ -4 \\ -4 \end{pmatrix} + 2 * \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix}$ ⇐

$d(x_4) = \begin{vmatrix} 1 & 5 & 4 \end{vmatrix} \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix} = 0$ ⇒ Update Weight $\underline{W} = \begin{pmatrix} 10 \\ -2 \\ 0 \end{pmatrix} - 2 * \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$ ⇐

$d(x_5) = \begin{vmatrix} 1 & 6 & 5 \end{vmatrix} \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix} = -104$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$

$d(x_6) = \begin{vmatrix} 1 & 6 & 6 \end{vmatrix} \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix} = -112$ ⇒ Keep same Weight $\underline{W} = \begin{pmatrix} 8 \\ -12 \\ -8 \end{pmatrix}$